

# Contour — the measurement picks the structure

`vrp_ratio < 1.30` → **NO\_TRADE** implied is not rich enough to sell

`skew_z ≥ +0.8` → **PUT\_CREDIT\_SPREAD** puts rich – sell puts, not cheap calls

`skew_z ≤ -0.8` → **CALL\_CREDIT\_SPREAD** calls rich – sell calls, not cheap puts

`otherwise` → **IRON\_CONDOR** both sides fair – sell both

## AI LOGIC — THE LEASH

Every wired model output can only make the agent trade **less**. Not a promise — a property of the import graph.

### THE MODEL MAY

- Name event windows to stand down in
- Veto a proposed structure
- Stand the whole book down

### THE MODEL MAY NEVER

- Choose a strike
- **Size** or price a position
- Reverse or widen anything

## ALPACA INFRASTRUCTURE

- **Orders go through the CLI, and that is a finding.** The MCP server cannot place multi-leg orders — the `legs` array arrives as a JSON string and fails validation. `alpaca-mcp-server` #97, open since July. The CLI places the identical 4-leg order correctly; our fix is upstream as #118.
- Reads are `alpaca-py`: chain snapshots carry Greeks but no `open_interest`, so they are merged with Trading API contract objects.
- Entries are a **three-rung limit ladder** from mid toward the bid, never a market order. `reconcile()` reads actual per-leg fills: paper issues random partials, and trusting the request puts legs out of ratio.
- The journal is an **append-only SHA-256 hash chain**. The dashboard recomputes it in your browser and prints the same verdict the CLI does.
- Autonomy is GitHub Actions: pre-open planning, a cycle every 15 minutes, a scheduled Thursday flatten. **No `pull_request` trigger** on the credentialed workflow.
- **No resting stop exists on a multi-leg position** — so the options book polls. A single equity leg *can* rest one, and the directional sleeve parks its stop GTC at the broker: the only exit that works overnight.

## MARKET ANALYSIS

The volatility premium is real, and it is **not uniform**. That dispersion is the entire opportunity.

SPY

1.55

paid to sell

QQQ

1.23

not paid enough

IWM

1.17

not paid enough

- Short premium is the crowded retail options trade — and the crowd sells **one structure regardless of the surface**.
- **Friction decides the universe, not opinion.** Single-name weeklies cost \$40–80 round trip against a \$30–42 modelled edge — it loses to costs before it starts. Three ETFs cost \$8–20.
- Alpaca serves **no earnings-date endpoint on any plan**, so a single-name design hangs its most important gate on data that does not exist.

# COMPETITIVE ANALYSIS

| SUBMISSION        | ITS OWN DESCRIPTION                                                                                             |
|-------------------|-----------------------------------------------------------------------------------------------------------------|
| Horizon Blackline | LLM proposes, deterministic risk gates authorize, hash-chained and auditable                                    |
| VRP Engine        | Harvests the variance risk premium with defined-risk spreads and risk gates                                     |
| AEGIS-Q           | Bounded AI selects a pre-validated bullish or bearish spread — or abstains                                      |
| EdgeStack         | Journals every trade and every refusal                                                                          |
| Contour           | Chooses which of four structures to sell, from measured 25-delta skew. And is runnable without our credentials. |

“LLM proposes, gates dispose, everything journaled” is table stakes now

```
python -m contour --replay
```

PERFORMANCE, ATTRIBUTED

The criterion asks for the P&L of **the submitted agent**. This account holds two traders, and the broker records which is which. 2026-09-04 04:01 UTC

| PLACED BY                                      | P&L       | OF START NAV |
|------------------------------------------------|-----------|--------------|
| The agent — every id it chose                  | +\$147.55 | +0.15%       |
| The operator — three discretionary tail trades | -\$642.00 | -0.64%       |
| Account total                                  | -\$494.45 | -0.49%       |

**Not our bookkeeping — a field the broker stamps.** Entries are prefixed contour- by loop.py; exits are named after the entry they close. Nothing else in the account is. A symbol touched by both is charged to the operator — the published number is the pessimistic one.

## REVENUE MODEL

### The audit layer is the product

The gate engine and the hash-chained decision record, licensed to brokers and RIAs who have to **defend** an automated decision after the fact. That is the durable asset here — not the alpha, which decays.

### Own capital

P&L is the revenue and no registration question arises. The honest ceiling on defined-risk premium selling is the credit, so this scales with capital, not with claims — and where we stepped outside that ceiling, the write-up names the trade and its negative expectancy.

**Not** signal subscriptions. Selling trade recommendations is investment advice and needs registration — saying so out loud is a credibility position, not a limitation.

## ROADMAP

### SHIPPED

- 3 ETFs, one locked expiry, 15-minute cycle
- **A \$30k long-QQQ sleeve** beside the options book — variance, not edge — funded *out of* the same -4% floor, not beside it
- **A \$4.4k long-call tail**, added *and closed* 2026-09-01 — the one negative-EV trade here, labelled as such. Sold for \$3,113, realising **-\$1,320**: the entire drawdown. A 387-cycle backtest found no edge worth paying ~15% over fair value to lever
- **A \$1.1k TQQQ call tail**, placed on instruction after the evidence against it was recorded — a gap-down bounce tests as noise ( $t = +0.42$ ). It breaks no gate, but used the last room in front of the floor: the entry ramp is *closed to zero* for the rest of the contest
- Twelve gates plus seven, 298 tests, chain verified in CI
- Record / replay: a fixture reruns the pipeline with no credentials
- Sizing from three published trend systems, after an audit caught the model anchoring at 0.5 for sixteen straight cycles

### NEXT

- **Trend-aware structure selection.** The map breaks ties toward the condor — which sells calls into a confirmed uptrend. Three trend systems say that is the wrong default; skew alone should not decide
- Expiry laddering and rolls
- Skew priors **learned** per underlying instead of hard-coded
- **More backtest history.** The harness is built and has run — 387 cycles against real historical option prices, importing the agent's own selection and gate code. Alpaca's history starts 2024-01-18, so 2.5 years is the entire available sample
- Portfolio vega and gamma caps, not per-position max loss alone
- Paid feed to close the indicative-vs-NBBO gap